

HIGHER ORDER DERIVATIVES

	Function	1 st Derivative	2 nd Derivative	3 rd Derivative	n th Derivative
Lagrange	$f(x)$				
Leibniz	y				

Example 1: Determine the 2nd derivative of $f(x) = x^4 + 3x^2 - 5$.

Example 2: Determine $\frac{d^2 y}{dx^2}$ given that $y = \frac{3x-5}{2x}$.

Example 3: Determine $f'''(x)$ of the following functions.

a) $f(x) = \frac{1}{x}$

b) $f(x) = 4x^7 + 5x^6 - 8x^4 + 23x - \frac{\pi}{3}$

c) $f(x) = \sqrt{x}$

Velocity and Acceleration

- **Position**, $s(t)$, is the location of an object at time, t .
- **Velocity**, $v(t)$, is the rate of change of position over time, so $v(t) = s'(t) = \frac{ds}{dt}$.
- **Acceleration**, $a(t)$, is the rate of change of velocity over time, so $a(t) = v'(t) = s''(t)$

$$\text{or } a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

Example 1: The height of a soccer ball above the ground at time, t after it is kicked into the air, is given by the formula

$$h(t) = -5t^2 + 30t + 1$$

where h is the height in metres, t is the time in seconds, and $t \geq 0$.

- Determine an equation that will calculate the instantaneous rate of change of the height of the ball with respect to time, t .
- Determine the velocity of the ball 0.5 seconds after it is kicked. At what other time will the ball have the same speed?
- At what time will the ball reach its maximum height? What is the maximum height reached by the ball?

- d) What is the instantaneous velocity of the ball as it hits the ground?

Example 2: A freight train leaves a train station and travels due north on a straight track. After t hours, the train is $s(t) = 18t^2 - 2t^3, 0 \leq t \leq 9$ kilometres north of the train station .

- a) Determine an expression for the velocity of the train at any time $0 \leq t \leq 9$.
- b) Determine the velocity of the train after 3 hours and after 7 hours.
- c) Determine the acceleration of the train after 1 hour.

Additional Questions

1. Given $f(x) = ax^2 + bx + c$ and $f(1) = 8$, $f'(1) = 3$ and $f''(1) = -4$, determine the values of a , b and c .
2. The position of a particle is given by $s(t) = 2t^3 - 39t^2 + 216t$, $t \geq 0$. Determine the acceleration of the particle at the instant when the velocity is zero.